

Fraction Operations

Use the cue, then fade the cue. Look at the cue card for 60 seconds. Then cover it before answering Round 1. Check the card only after you commit to an answer.

CUE FOCUS Add or subtract: get a common denominator, then combine numerators ($\frac{1}{4} + \frac{2}{4} = \frac{3}{4}$). Multiply: multiply across ($\frac{2}{3} \times \frac{3}{5} = \frac{6}{15} = \frac{2}{5}$). Divide: Keep-Change-Flip ($\frac{1}{2} \div \frac{1}{4} = \frac{1}{2} \times \frac{4}{1} = 2$). Simplify by the GCF.

Round 1 — Retrieve It

1. What must two fractions share before you can add or subtract them?

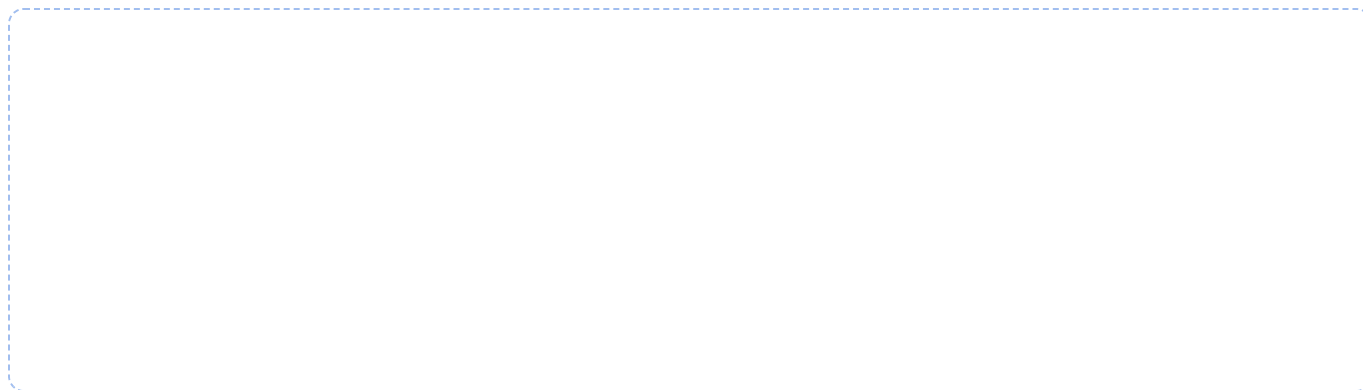
2. How do you multiply two fractions?

3. What do the three steps of Keep-Change-Flip mean when dividing fractions?

4. How do you simplify a fraction to lowest terms?

Round 2 — Sketch It

Draw two same-size rectangles split into fourths. Shade $\frac{1}{4}$ in one and $\frac{2}{4}$ in the other to show $\frac{1}{4} + \frac{2}{4}$. Circle the total shaded amount and write it as a fraction.



Fraction Operations — Apply It

Keep the cue card covered. Solve first. Then use the cue to check, revise, and explain what you changed.

Round 3 — Apply the Relationship

1. Find $\frac{1}{4} + \frac{2}{4}$.

2. Find $\frac{2}{3} \times \frac{3}{5}$ and simplify.

3. Find $\frac{1}{2} \div \frac{1}{4}$.

Round 4 — Reason from Evidence

A student writes $\frac{1}{2} + \frac{1}{3} = \frac{2}{5}$ by adding the tops and adding the bottoms. Explain why that breaks the common-denominator rule, then give the correct answer.

Fraction Operations — ACE It

Use the cue card only if you need it. The goal is to show what you can explain, connect, and use on your own.

A**Articulate**

Explain why adding fractions needs a common denominator but multiplying fractions does not. Use one example that is not printed on the cue card.

C**Connect**

Connect Keep-Change-Flip to multiplication. Why does dividing by $\frac{1}{4}$ give the same result as multiplying by $\frac{4}{1}$?

E**Extend**

Create your own fraction division problem. Show the work using Keep-Change-Flip and simplify the answer by the GCF.

Confidence check. Circle one after you finish: I still need the cue / I need it once in a while / I can explain this without it.

Teacher Key — Fraction Operations

Remove this page before student copying. Answers below give expected ideas, not required wording.

Retrieve It

1. A common denominator.
2. Multiply across: numerator $\times$ numerator and denominator $\times$ denominator.
3. Keep the first fraction, change $\div$ to $\times$, and flip the second fraction to its reciprocal.
4. Divide the numerator and denominator by their greatest common factor (GCF).

Sketch It — look for

Two fourths-partitioned rectangles: $1/4$ shaded in one and $2/4$ shaded in the other, with the combined total circled and written as **$3/4$** .

Apply It

1. $1/4 + 2/4 = \mathbf{3/4}$ (denominators already match).
2. $2/3 \times 3/5 = 6/15 = \mathbf{2/5}$ (divide by GCF 3).
3. $1/2 \div 1/4 = 1/2 \times 4/1 = 4/2 = \mathbf{2}$.

Reason from Evidence — look for

You cannot add tops and bottoms; the fractions need a common denominator first. Using twelfths: $1/2 = 3/6$ and $1/3 = 2/6$, so $1/2 + 1/3 = 3/6 + 2/6 = \mathbf{5/6}$, not $2/5$.

ACE — Extend look-for

Many answers possible. Check that the problem divides fractions, the work uses Keep-Change-Flip, and the answer is simplified by the GCF.

Suggested use. Run Page 1 with the cue card available, Page 2 with the cue mostly covered, and Page 3 as independent reflection. Let students uncover the cue to self-correct in a different color or with a revision mark.